

The University of the State of New York
REGENTS HIGH SCHOOL EXAMINATION

ALGEBRA II (Common Core)

Wednesday, June 1, 2016 — 9:15 a.m. to 12:15 p.m., only

Student Name: _____

School Name: _____

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

Print your name and the name of your school on the lines above.

A separate answer sheet for Part I has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 37 questions. You must answer all questions in this examination. Record your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in Parts II, III, and IV directly in this booklet. All work should be written in pen, except graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale.

The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will *not* be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

Notice...

A graphing calculator and a straightedge (ruler) must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

High School Math Reference Sheet

1 inch = 2.54 centimeters	1 kilometer = 0.62 mile	1 cup = 8 fluid ounces
1 meter = 39.37 inches	1 pound = 16 ounces	1 pint = 2 cups
1 mile = 5280 feet	1 pound = 0.454 kilogram	1 quart = 2 pints
1 mile = 1760 yards	1 kilogram = 2.2 pounds	1 gallon = 4 quarts
1 mile = 1.609 kilometers	1 ton = 2000 pounds	1 gallon = 3.785 liters
		1 liter = 0.264 gallon
		1 liter = 1000 cubic centimeters

Triangle	$A = \frac{1}{2}bh$
Parallelogram	$A = bh$
Circle	$A = \pi r^2$
Circle	$C = \pi d$ or $C = 2\pi r$
General Prisms	$V = Bh$
Cylinder	$V = \pi r^2 h$
Sphere	$V = \frac{4}{3}\pi r^3$
Cone	$V = \frac{1}{3}\pi r^2 h$
Pyramid	$V = \frac{1}{3}Bh$

Pythagorean Theorem	$a^2 + b^2 = c^2$
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Arithmetic Sequence	$a_n = a_1 + (n - 1)d$
Geometric Sequence	$a_n = a_1 r^{n-1}$
Geometric Series	$S_n = \frac{a_1 - a_1 r^n}{1 - r}$ where $r \neq 1$
Radians	1 radian = $\frac{180}{\pi}$ degrees
Degrees	1 degree = $\frac{\pi}{180}$ radians
Exponential Growth/Decay	$A = A_0 e^{k(t - t_0)} + B_0$

Part I

Answer all 24 questions in this part. Each correct answer will receive 2 credits. No partial credit will be allowed. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. Record your answers on your separate answer sheet. [48]

Use this space for
computations.

- 1 When $b > 0$ and d is a positive integer, the expression $(3b)^{\frac{2}{d}}$ is equivalent to

(1) $\frac{1}{(\sqrt[d]{3b})^2}$

(3) $\frac{1}{\sqrt{3b^d}}$

(2) $(\sqrt{3b})^d$

(4) $(\sqrt[d]{3b})^2$

- 2 Julie averaged 85 on the first three tests of the semester in her mathematics class. If she scores 93 on each of the remaining tests, her average will be 90. Which equation could be used to determine how many tests, T , are left in the semester?

(1) $\frac{255 + 93T}{3T} = 90$

(3) $\frac{255 + 93T}{T + 3} = 90$

(2) $\frac{255 + 90T}{3T} = 93$

(4) $\frac{255 + 90T}{T + 3} = 93$

- 3 Given i is the imaginary unit, $(2 - yi)^2$ in simplest form is

(1) $y^2 - 4yi + 4$

(3) $-y^2 + 4$

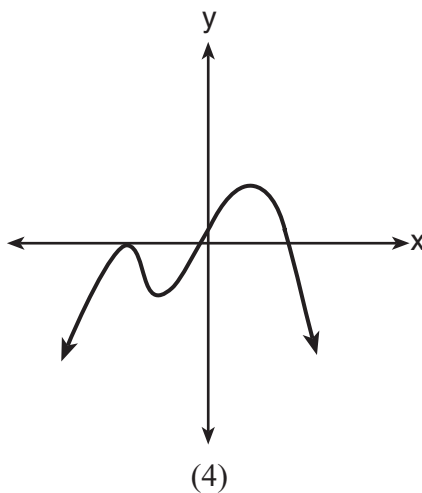
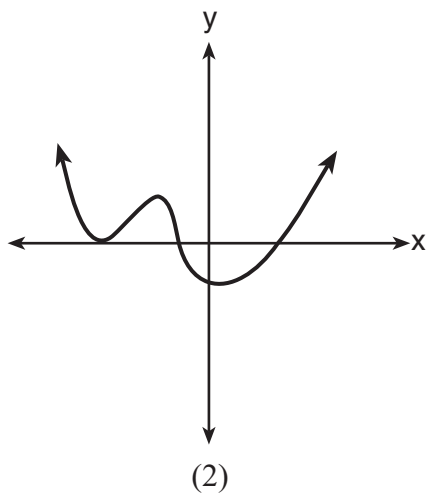
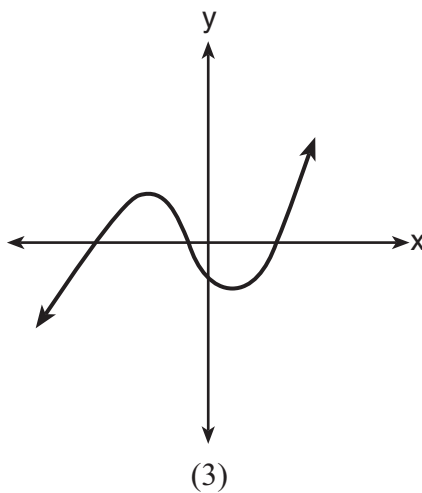
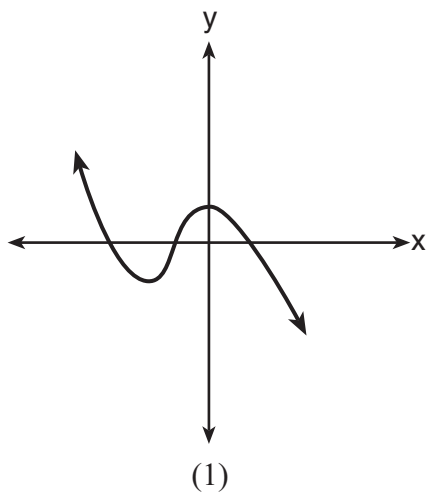
(2) $-y^2 - 4yi + 4$

(4) $y^2 + 4$

Use this space for
computations.

4 Which graph has the following characteristics?

- three real zeros
- as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$
- as $x \rightarrow \infty$, $f(x) \rightarrow \infty$



5 The solution set for the equation $\sqrt{56 - x} = x$ is

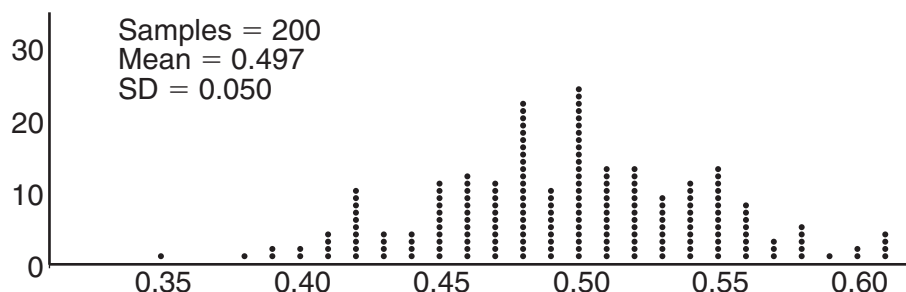
- | | |
|-----------------|-------------|
| (1) $\{-8, 7\}$ | (3) $\{7\}$ |
| (2) $\{-7, 8\}$ | (4) $\{\}$ |

**Use this space for
computations.**

6 The zeros for $f(x) = x^4 - 4x^3 - 9x^2 + 36x$ are

- (1) $\{0, \pm 3, 4\}$ (3) $\{0, \pm 3, -4\}$
 (2) $\{0, 3, 4\}$ (4) $\{0, 3, -4\}$

7 Anne has a coin. She does not know if it is a fair coin. She flipped the coin 100 times and obtained 73 heads and 27 tails. She ran a computer simulation of 200 samples of 100 fair coin flips. The output of the proportion of heads is shown below.



Given the results of her coin flips and of her computer simulation, which statement is most accurate?

- (1) 73 of the computer's next 100 coin flips will be heads.
 (2) 50 of her next 100 coin flips will be heads.
 (3) Her coin is not fair.
 (4) Her coin is fair.

8 If $g(c) = 1 - c^2$ and $m(c) = c + 1$, then which statement is *not* true?

- (1) $g(c) \cdot m(c) = 1 + c - c^2 - c^3$
 (2) $g(c) + m(c) = 2 + c - c^2$
 (3) $m(c) - g(c) = c + c^2$
 (4) $\frac{m(c)}{g(c)} = \frac{-1}{1 - c}$

**Use this space for
computations.**

- 9** The heights of women in the United States are normally distributed with a mean of 64 inches and a standard deviation of 2.75 inches. The percent of women whose heights are between 64 and 69.5 inches, to the *nearest whole percent*, is

- | | |
|--------|--------|
| (1) 6 | (3) 68 |
| (2) 48 | (4) 95 |

- 10** The formula below can be used to model which scenario?

$$\begin{aligned}a_1 &= 3000 \\ a_n &= 0.80a_{n-1}\end{aligned}$$

- (1) The first row of a stadium has 3000 seats, and each row thereafter has 80 more seats than the row in front of it.
- (2) The last row of a stadium has 3000 seats, and each row before it has 80 fewer seats than the row behind it.
- (3) A bank account starts with a deposit of \$3000, and each year it grows by 80%.
- (4) The initial value of a specialty toy is \$3000, and its value each of the following years is 20% less.

- 11** Sean's team has a baseball game tomorrow. He pitches 50% of the games. There is a 40% chance of rain during the game tomorrow. If the probability that it rains given that Sean pitches is 40%, it can be concluded that these two events are

- | | |
|-----------------|------------------------|
| (1) independent | (3) mutually exclusive |
| (2) dependent | (4) complements |

Use this space for
computations.

12 A solution of the equation $2x^2 + 3x + 2 = 0$ is

(1) $-\frac{3}{4} + \frac{1}{4}i\sqrt{7}$

(3) $-\frac{3}{4} + \frac{1}{4}\sqrt{7}$

(2) $-\frac{3}{4} + \frac{7}{4}i$

(4) $\frac{1}{2}$

13 The Ferris wheel at the landmark Navy Pier in Chicago takes 7 minutes to make one full rotation. The height, H , in feet, above the ground of one of the six-person cars can be modeled by

$H(t) = 70 \sin\left(\frac{2\pi}{7}(t - 1.75)\right) + 80$, where t is time, in minutes. Using

$H(t)$ for one full rotation, this car's minimum height, in feet, is

(1) 150

(3) 10

(2) 70

(4) 0

14 The expression $\frac{4x^3 + 5x + 10}{2x + 3}$ is equivalent to

(1) $2x^2 + 3x - 7 + \frac{31}{2x + 3}$

(3) $2x^2 + 2.5x + 5 + \frac{15}{2x + 3}$

(2) $2x^2 - 3x + 7 - \frac{11}{2x + 3}$

(4) $2x^2 - 2.5x - 5 - \frac{20}{2x + 3}$

15 Which function represents exponential decay?

(1) $y = 2^{0.3t}$

(3) $y = \left(\frac{1}{2}\right)^{-t}$

(2) $y = 1.2^{3t}$

(4) $y = 5^{-t}$

Use this space for
computations.

16 Given $f^{-1}(x) = -\frac{3}{4}x + 2$, which equation represents $f(x)$?

(1) $f(x) = \frac{4}{3}x - \frac{8}{3}$

(3) $f(x) = \frac{3}{4}x - 2$

(2) $f(x) = -\frac{4}{3}x + \frac{8}{3}$

(4) $f(x) = -\frac{3}{4}x + 2$

17 A circle centered at the origin has a radius of 10 units. The terminal side of an angle, θ , intercepts the circle in Quadrant II at point C . The y -coordinate of point C is 8. What is the value of $\cos \theta$?

(1) $-\frac{3}{5}$

(3) $\frac{3}{5}$

(2) $-\frac{3}{4}$

(4) $\frac{4}{5}$

18 Which statement about the graph of $c(x) = \log_6 x$ is *false*?

(1) The asymptote has equation $y = 0$.

(2) The graph has no y -intercept.

(3) The domain is the set of positive reals.

(4) The range is the set of all real numbers.

19 The equation $4x^2 - 24x + 4y^2 + 72y = 76$ is equivalent to

(1) $4(x - 3)^2 + 4(y + 9)^2 = 76$

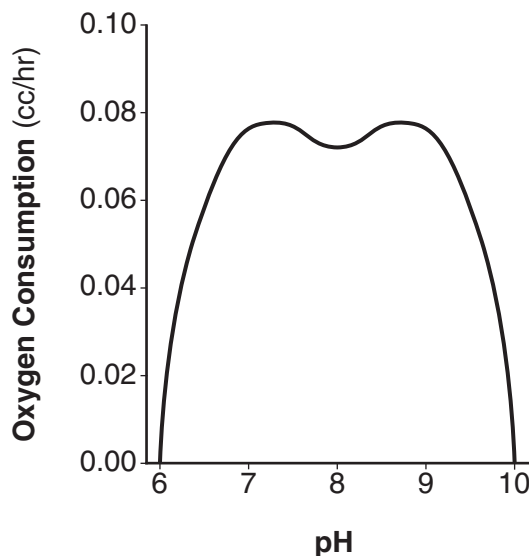
(2) $4(x - 3)^2 + 4(y + 9)^2 = 121$

(3) $4(x - 3)^2 + 4(y + 9)^2 = 166$

(4) $4(x - 3)^2 + 4(y + 9)^2 = 436$

Use this space for
computations.

- 20** There was a study done on oxygen consumption of snails as a function of pH, and the result was a degree 4 polynomial function whose graph is shown below.



Which statement about this function is *incorrect*?

- (1) The degree of the polynomial is even.
- (2) There is a positive leading coefficient.
- (3) At two pH values, there is a relative maximum value.
- (4) There are two intervals where the function is decreasing.

- 21** Last year, the total revenue for Home Style, a national restaurant chain, increased 5.25% over the previous year. If this trend were to continue, which expression could the company's chief financial officer use to approximate their monthly percent increase in revenue? [Let m represent months.]

- (1) $(1.0525)^m$
- (2) $(1.0525)^{\frac{12}{m}}$
- (3) $(1.00427)^m$
- (4) $(1.00427)^{\frac{m}{12}}$

**Use this space for
computations.**

22 Which value, to the *nearest tenth*, is *not* a solution of $p(x) = q(x)$ if $p(x) = x^3 + 3x^2 - 3x - 1$ and $q(x) = 3x + 8$?

- (1) -3.9 (3) 2.1
(2) -1.1 (4) 4.7

23 The population of Jamesburg for the years 2010 – 2013, respectively, was reported as follows:

250,000 250,937 251,878 252,822

How can this sequence be recursively modeled?

- (1) $j_n = 250,000(1.00375)^{n-1}$
(2) $j_n = 250,000 + 937^{(n-1)}$
(3) $j_1 = 250,000$
 $j_n = 1.00375j_{n-1}$
(4) $j_1 = 250,000$
 $j_n = j_{n-1} + 937$

24 The voltage used by most households can be modeled by a sine function. The maximum voltage is 120 volts, and there are 60 cycles *every second*. Which equation best represents the value of the voltage as it flows through the electric wires, where t is time in seconds?

- (1) $V = 120 \sin(t)$ (3) $V = 120 \sin(60\pi t)$
(2) $V = 120 \sin(60t)$ (4) $V = 120 \sin(120\pi t)$
-

Part II

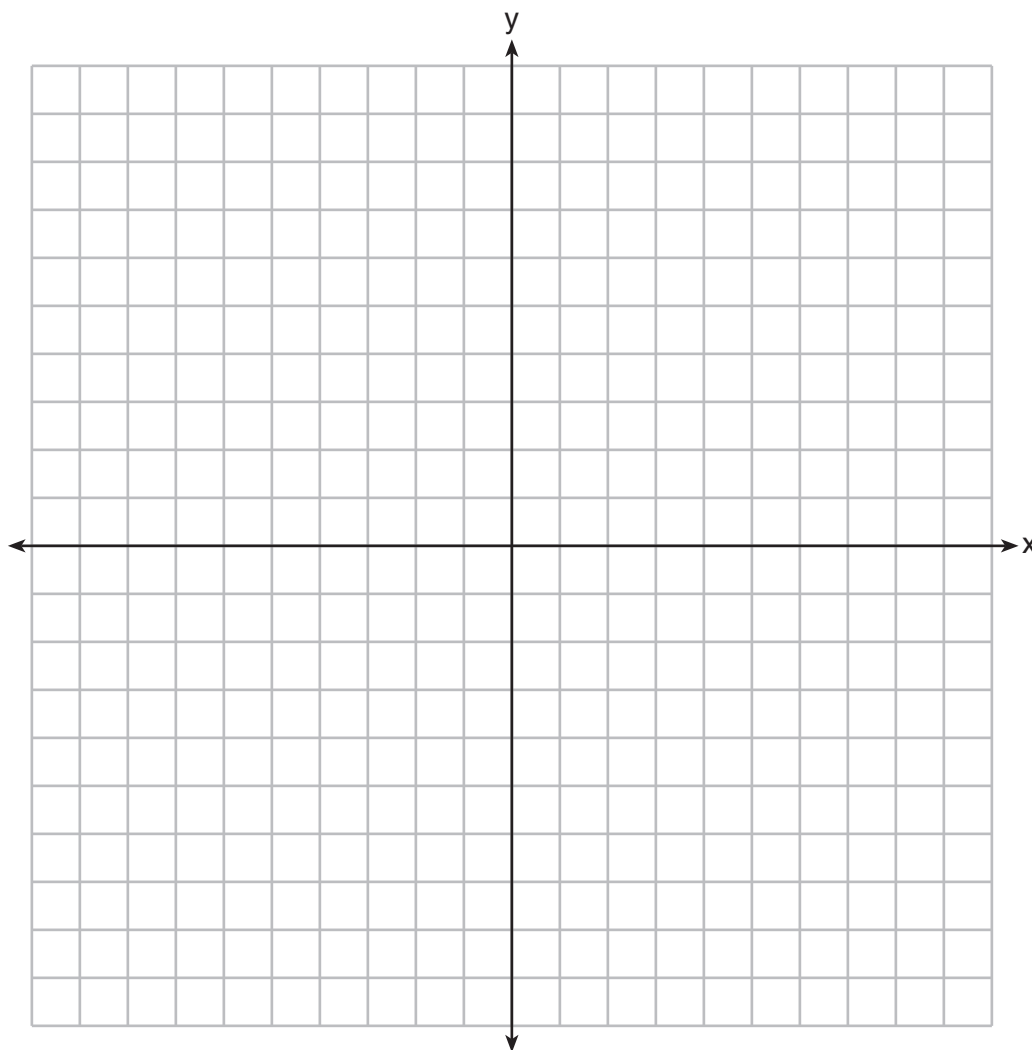
Answer all 8 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [16]

25 Solve for x : $\frac{1}{x} - \frac{1}{3} = -\frac{1}{3x}$

26 Describe how a controlled experiment can be created to examine the effect of ingredient X in a toothpaste.

27 Determine if $x - 5$ is a factor of $2x^3 - 4x^2 - 7x - 10$. Explain your answer.

- 28** On the axes below, graph *one* cycle of a cosine function with amplitude 3, period $\frac{\pi}{2}$, midline $y = -1$, and passing through the point $(0,2)$.



29 A suburban high school has a population of 1376 students. The number of students who participate in sports is 649. The number of students who participate in music is 433. If the probability that a student participates in either sports or music is $\frac{974}{1376}$, what is the probability that a student participates in both sports and music?

- 30** The directrix of the parabola $12(y + 3) = (x - 4)^2$ has the equation $y = -6$. Find the coordinates of the focus of the parabola.

31 Algebraically prove that $\frac{x^3 + 9}{x^3 + 8} = 1 + \frac{1}{x^3 + 8}$, where $x \neq -2$.

32 A house purchased 5 years ago for \$100,000 was just sold for \$135,000. Assuming exponential growth, approximate the annual growth rate, to the *nearest percent*.

Part III

Answer all 4 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [16]

33 Solve the system of equations shown below algebraically.

$$(x - 3)^2 + (y + 2)^2 = 16$$

$$2x + 2y = 10$$

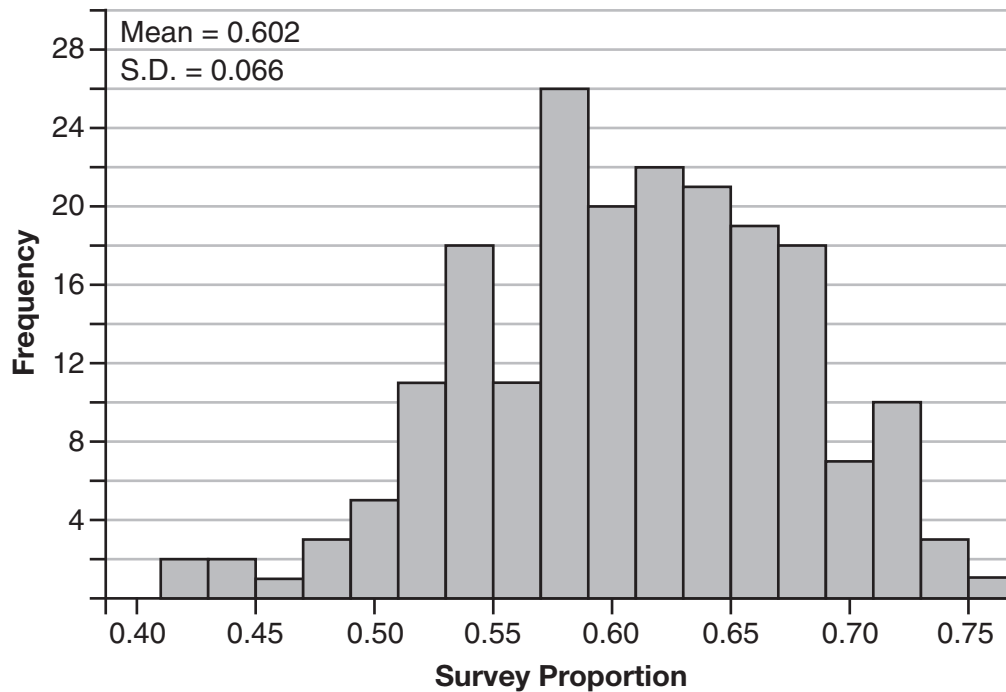
34 Alexa earns \$33,000 in her first year of teaching and earns a 4% increase in each successive year.

Write a geometric series formula, S_n , for Alexa's total earnings over n years.

Use this formula to find Alexa's total earnings for her first 15 years of teaching, to the *nearest cent*.

- 35** Fifty-five students attending the prom were randomly selected to participate in a survey about the music choice at the prom. Sixty percent responded that a DJ would be preferred over a band. Members of the prom committee thought that the vote would have 50% for the DJ and 50% for the band.

A simulation was run 200 times, each of sample size 55, based on the premise that 60% of the students would prefer a DJ. The approximate normal simulation results are shown below.



Using the results of the simulation, determine a plausible interval containing the middle 95% of the data. Round all values to the *nearest hundredth*.

Members of the prom committee are concerned that a vote of all students attending the prom may produce a 50% – 50% split. Explain what statistical evidence supports this concern.

36 Which function shown below has a greater average rate of change on the interval $[-2, 4]$? Justify your answer.

x	f(x)
-4	0.3125
-3	0.625
-2	1.25
-1	2.5
0	5
1	10
2	20
3	40
4	80
5	160
6	320

$$g(x) = 4x^3 - 5x^2 + 3$$

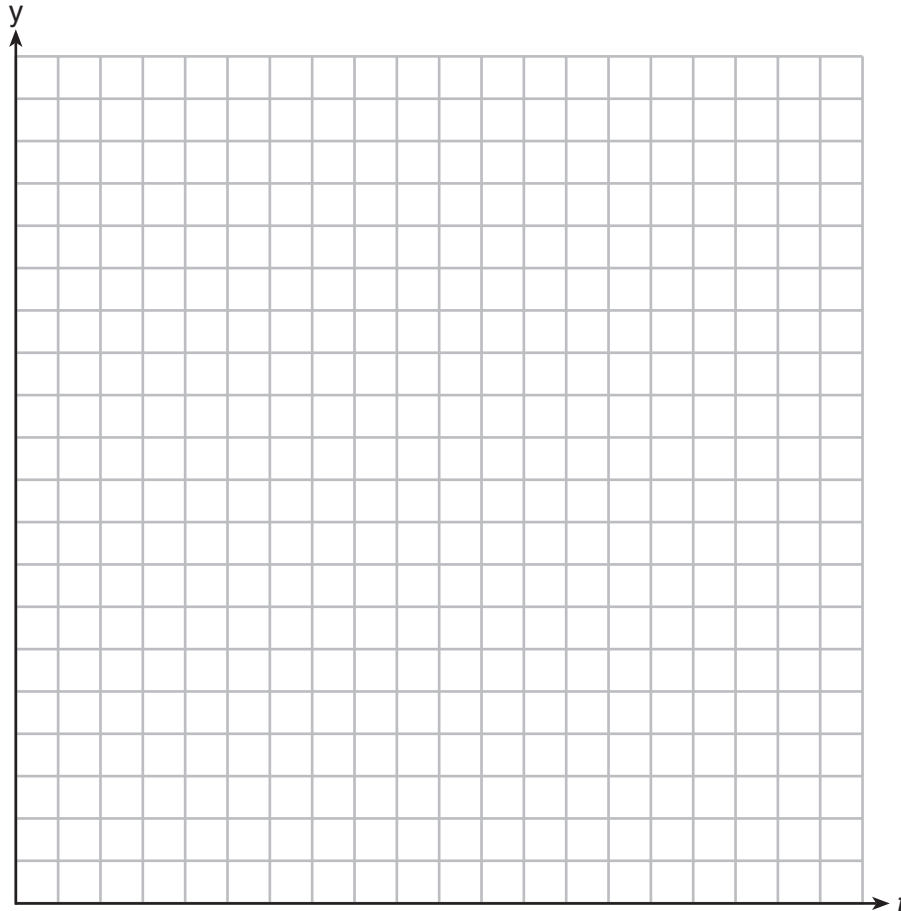
Part IV

Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided to determine your answer. Note that diagrams are not necessarily drawn to scale. A correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [6]

- 37 Drugs break down in the human body at different rates and therefore must be prescribed by doctors carefully to prevent complications, such as overdosing. The breakdown of a drug is represented by the function $N(t) = N_0(e)^{-rt}$, where $N(t)$ is the amount left in the body, N_0 is the initial dosage, r is the decay rate, and t is time in hours. Patient A, $A(t)$, is given 800 milligrams of a drug with a decay rate of 0.347. Patient B, $B(t)$, is given 400 milligrams of another drug with a decay rate of 0.231.

Write two functions, $A(t)$ and $B(t)$, to represent the breakdown of the respective drug given to each patient.

Graph each function on the set of axes below.



To the *nearest hour*, t , when does the amount of the given drug remaining in patient B begin to exceed the amount of the given drug remaining in patient A ?

The doctor will allow patient A to take another 800 milligram dose of the drug once only 15% of the original dose is left in the body. Determine, to the *nearest tenth of an hour*, how long patient A will have to wait to take another 800 milligram dose of the drug.

If the student's responses for the multiple-choice questions are being hand scored prior to being scanned, the scorer must be careful not to make any marks on the answer sheet except to record the scores in the designated score boxes. Marks elsewhere on the answer sheet will interfere with the accuracy of the scanning.

Part I

Allow a total of 48 credits, 2 credits for each of the following.

(1) 4	(9) 2	(17) 1
(2) 3	(10) 4	(18) 1
(3) 2	(11) 1	(19) 4
(4) 3	(12) 1	(20) 2
(5) 3	(13) 3	(21) 3
(6) 1	(14) 2	(22) 4
(7) 3	(15) 4	(23) 3
(8) 4	(16) 2	(24) 4

Updated information regarding the rating of this examination may be posted on the New York State Education Department's web site during the rating period. Check this web site at: <http://www.p12.nysed.gov/assessment/> and select the link "Scoring Information" for any recently posted information regarding this examination. This site should be checked before the rating process for this examination begins and several times throughout the Regents Examination period.

The Department is providing supplemental scoring guidance, the "Model Response Set," for the Regents Examination in Algebra II (Common Core). This guidance is recommended to be part of the scorer training. Schools are encouraged to incorporate the Model Response Sets into the scorer training or to use them as additional information during scoring. While not reflective of all scenarios, the model responses selected for the Model Response Set illustrate how less common student responses to constructed-response questions may be scored. The Model Response Set will be available on the Department's web site at <http://www.nysedregents.org/algebratwo/>.

**Map to the Common Core Learning Standards
Algebra II (Common Core)
June 2016**

Question	Type	Credits	Cluster
1	Multiple Choice	2	N-RN.A
2	Multiple Choice	2	A-CED.A
3	Multiple Choice	2	N-CN.A
4	Multiple Choice	2	F-IF.C
5	Multiple Choice	2	A-REI.A
6	Multiple Choice	2	A-APR.B
7	Multiple Choice	2	S-IC.A
8	Multiple Choice	2	F-BF.A
9	Multiple Choice	2	S-ID.A
10	Multiple Choice	2	F-BF.A
11	Multiple Choice	2	S-CP.A
12	Multiple Choice	2	N-CN.C
13	Multiple Choice	2	F-IF.B
14	Multiple Choice	2	A-APR.D
15	Multiple Choice	2	F-IF.C
16	Multiple Choice	2	F-BF.B
17	Multiple Choice	2	F-TF.A
18	Multiple Choice	2	F-IF.C
19	Multiple Choice	2	A-SSE.A
20	Multiple Choice	2	F-IF.B

21	Multiple Choice	2	A-SSE.B
22	Multiple Choice	2	A-REI.D
23	Multiple Choice	2	F-BF.A
24	Multiple Choice	2	F-TF.B
25	Constructed Response	2	A-REI.A
26	Constructed Response	2	S-IC.B
27	Constructed Response	2	A-APR.B
28	Constructed Response	2	F-IF.C
29	Constructed Response	2	S-CP.B
30	Constructed Response	2	G-GPE.A
31	Constructed Response	2	A-SSE.A
32	Constructed Response	2	F-LE.A
33	Constructed Response	4	A-REI.C
34	Constructed Response	4	A-SSE.B
35	Constructed Response	4	S-IC.B
36	Constructed Response	4	F-IF.C
37	Constructed Response	6	A-REI.D

Part II

For each question, use the specific criteria to award a maximum of 2 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(25) [2] 4, and correct work is shown.

[1] Appropriate work is shown, but one computational error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] 4, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(26) [2] A correct description of a controlled experiment is written, such as indicating two randomly assigned groups, one with ingredient X and one without ingredient X .

[1] One conceptual error is made.

or

[1] An incomplete description of a controlled experiment is written.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(27) [2] No, and correct work is shown, and a correct explanation is written.

[1] Appropriate work is shown, but one computational error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] Correct work is shown, but no explanation or an incorrect explanation is written.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(28) [2] A correct graph is drawn.

[1] One graphing error is made.

or

[1] One conceptual error is made, such as graphing more than one cycle.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(29) [2] $\frac{108}{1376}$ or an equivalent fraction, and correct work is shown.

[1] Appropriate work is shown, but one computational error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] $\frac{108}{1376}$, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(30) [2] (4,0), and correct work is shown.

[1] Appropriate work is shown, but one computational error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] (4,0), but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(31) [2] A correct algebraic proof is shown.

[1] Appropriate work is shown, but one computational or simplification error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(32) [2] 6, and correct work is shown.

[1] Appropriate work is shown, but one computational or rounding error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] 6, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

Part III

For each question, use the specific criteria to award a maximum of 4 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(33) [4] $(7, -2)$ and $(3, 2)$ or equivalent solutions, and correct algebraic work is shown.

[3] Appropriate work is shown, but one computational, factoring, or substitution error is made.

or

[3] Appropriate work is shown, but only one correct solution is found or only the x -values or y -values are found.

[2] Appropriate work is shown, but two or more computational, factoring, or substitution errors are made.

or

[2] Appropriate work is shown, but one conceptual error is made.

or

[2] A correct substitution into the quadratic formula is made, but no further correct work is shown.

or

[2] $(7, -2)$ and $(3, 2)$, but a method other than algebraic is used.

[1] Appropriate work is shown, but one conceptual error and one computational, factoring, or substitution error are made.

or

[1] A correct quadratic equation in one variable is written, but no further correct work is shown.

or

[1] $(7, -2)$ and $(3, 2)$, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(34) [4] $S_n = \frac{33,000 - 33,000(1.04)^n}{1 - 1.04}$ or an equivalent equation is written and 660,778.39, and correct work is shown.

[3] Appropriate work is shown, but one computational or simplification error is made.

or

[3] Appropriate work is shown, but one notation error is made, such as writing the expression $\frac{33,000 - 33,000(1.04)^n}{1 - 1.04}$, or not using n .

[2] Appropriate work is shown, but two or more computational, notation, or simplification errors are made.

or

[2] Appropriate work is shown, but one conceptual error is made.

or

[2] $S_n = \frac{33,000 - 33,000(1.04)^n}{1 - 1.04}$, but no further correct work is shown.

[1] Appropriate work is shown, but one conceptual error and one computational, notation, or simplification error are made.

or

[1] 660,778.39, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (35) [4] (0.47, 0.73) and correct work is shown, and a correct statistical explanation is written.
- [3] Appropriate work is shown, but one computational error is made.
- or***
- [3] Correct work is shown to find (0.47, 0.73), but the explanation is incomplete or nonstatistical.
- [2] Appropriate work is shown, but two or more computational errors are made.
- or***
- [2] Appropriate work is shown, but one conceptual error is made.
- or***
- [2] Appropriate work is shown to find (0.47, 0.73), but no further correct work is shown.
- or***
- [2] A correct statistical explanation is written, but no further correct work is shown.
- [1] Appropriate work is shown, but one conceptual error and one computational error are made.
- or***
- [1] (0.47, 0.73), but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (36) [4] g , and a correct justification is given.
- [3] Appropriate work is shown, but one computational error is made.
- or***
- [3] Correct rates of change are computed, but no function is indicated.
- [2] Appropriate work is shown, but two or more computational errors are made.
- or***
- [2] Appropriate work is shown, but one conceptual error is made.
- or***
- [2] Appropriate work is shown to find the rate of change for f is 13.125 or the rate of change for g is 38, but no further correct work is shown.
- [1] Appropriate work is shown, but one conceptual error and one computational error are made.
- or***
- [1] 13.125 and 38, but no work is shown.
- [0] 13.125 or 38, but the rates of change are not labeled, and no further correct work is shown.
- or***
- [0] g , but no work is shown.
- or***
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
-

Part IV

For this question, use the specific criteria to award a maximum of 6 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(37) [6] $A(t) = 800e^{-0.347t}$ and $B(t) = 400e^{-0.231t}$, correct graphs are drawn and at least one is labeled, 6, 5.5, and correct work is shown.

[5] Appropriate work is shown, but one computational, graphing, labeling, or rounding error is made.

[4] Appropriate work is shown, but two computational, graphing, labeling, or rounding errors are made.

or

[4] Appropriate work is shown, but one conceptual error is made.

[3] Appropriate work is shown, but three or more computational, graphing, labeling, or rounding errors are made.

or

[3] Appropriate work is shown, but one conceptual error and one computational, graphing, labeling, or rounding error are made.

[2] Appropriate work is shown, but two conceptual errors are made.

or

[2] Correct graphs are drawn and at least one is labeled, but no further correct work is shown.

or

[2] 5.5 and correct work is shown, but no further correct work is shown.

or

[2] 6 and 5.5, but no work is shown.

[1] Appropriate work is shown, but two conceptual errors and one computational, graphing, labeling, or rounding errors are made.

or

[1] $A(t) = 800e^{-0.347t}$ and $B(t) = 400e^{-0.231t}$, but no further correct work is shown.

or

[1] $A(t)$ or $B(t)$ is graphed correctly, but no further correct work is shown.

or

[1] 6 or 5.5, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.